LOGIXO TECH COMMERCIAL QUOTATION & SOW

YouZen PostgreSQL & Supabase Concurrency Optimization • Phase 1 Statement of Work

Client: Luca & YouZen Technical Team
Engagement: Phase 1 Performance Audit & Load Simulation
Technical Lead: Viranga Lakshan (Lead Solutions Architect)
Date: November 2026

> **Phase 1 Fixed-Price Scope of Work & Commercial Quotation**
> This document provides the formal fixed-price quotation, delivery timeline, itemized scope confirmation, and team responsibility breakdown for Phase 1 of the YouZen database optimization engagement, supporting 100 concurrent restaurants and 100 hair salons.

1. PHASE 1 COMMERCIAL QUOTATION & TERMS

Transparent Fixed-Price Deliverable with Zero Hidden Costs

ENGAGEMENT DELIVERABLE	SCOPE SUMMARY & TOOLING	AMOUNT (USD)
Phase 1: Full Audit & Staging Simulation	Baseline telemetry, synthetic data generator (100 venues each), independent & combined peak k6 load tests, lock/booking checks, and comprehensive findings report.	**$1,000.00**
Additional Infrastructure Costs	USD 0 under agreed existing staging-environment assumptions. Any required paid Supabase plan upgrades, additional cloud infrastructure or third-party services will be identified and approved by the client before being incurred.	**USD 0.00**
Delivery Timeline	**10 to 14 business days (2 to 3 weeks)**, matching the Phase 1 sprint schedule (Weeks 1–3) in our Technical Proposal.	**Fixed Timeline**
Payment Milestone Structure	Single milestone funded in Upwork escrow ($1,000) upon kickoff, released upon delivery of verified test scripts and report.	**100% Escrow Protected**

2. CONFIRMATION OF PHASE 1 SCOPE & TECHNICAL INCLUSIONS

All Requested Deliverables Fully Included in the $1,000 Fixed Price

1. Baseline Telemetry & Audit — TELEMETRY

Measuring existing query performance, API latencies (p50, p95, p99), connection pooler saturation in PgBouncer, lock wait duration trees, and buffer cache hit rates via `pg_stat_statements`. Tracing automatic client polling and Supabase Realtime replication overhead.

2. Synthetic Data Setup — STAGING HARNESS

Building an automated, realistic data seeding generator for 100 restaurants (tables, active menus, checks, line items) and 100 hair salons (stylists, working hours, services, booking windows) with multi-tenant relational integrity.

3. Independent App Load Tests — ISOLATED RUNS

- **Restaurant Rush:** 100 checkouts + 200 staff mobile phones concurrently submitting items, printing kitchen tickets, and settling checks.
- **Salon Booking Peak:** Multi-column stylist calendar rendering, availability slot lookups, and booking commits.

4. Combined Peak Concurrency Test — PEAK STRESS

Simultaneously executing the restaurant dinner rush and salon booking peak to measure cross-tenant database CPU spikes, lock queuing, and connection pool exhaustion under joint peak loads.

CONCURRENCY & BUSINESS RULE INTEGRITY VERIFICATION

Stress-Testing High-Risk Scheduling and POS Workflows

Conflicting Bookings Test — ZERO OVERLAPS

Simulating parallel booking requests attempting to claim the exact same stylist time slot simultaneously to physically verify that race conditions cannot cause double-bookings.

Valid Parallel Bookings — THROUGHPUT

Verifying that concurrent bookings across different stylists or non-overlapping time windows proceed simultaneously with zero lock blocking or artificial queuing.

Order Idempotency Test — ZERO DUPLICATES

Simulating staff mobile phone network retries during dinner rush to verify that duplicate kitchen tickets and redundant check settlements are completely prevented.

PHASE 1 FINAL HANDOVER DELIVERABLES

What YouZen Receives at the Conclusion of Phase 1

Reusable Test Scripts in Git — CODE ASSET

Complete, parameterized **k6** (JavaScript/TypeScript) load scripts handed over in your repository with synthetic seeders and a clean README, allowing your team to re-run automated benchmarks on demand.

Empirical Audit Report & Roadmap — TECHNICAL REPORT

A comprehensive findings report documenting baseline vs. peak latency percentiles (p50, p95, p99), requests per second (RPS), connection curves, and a prioritized, actionable backlog of fixes for Phase 2.

3. PHASE 2 INDICATIVE COMMERCIAL FRAMEWORK

Remediation Implementation & Benchmark Re-Verification

Phase 2 Budget & Scope Governance — CAPPED FIXED PRICE

Indicative Fixed-Price Range: $1,500 – $2,500 USD
Pricing Basis: Estimated from 30–50 hours at our standard senior engineering rate ($45–$50/hr), delivered strictly as a capped fixed-price milestone for complete cost certainty.
Estimated Timeline: 2 to 3 weeks (Weeks 4–6), matching Phase 2 in our Technical Proposal.

Scope & Governance: Implementing prioritized composite indexes, query rewrites, RLS policy refactoring, concurrency guards, and Realtime channel scoping, followed by a complete re-run of the k6 benchmark suite. The final milestone budget will be agreed after reviewing the Phase 1 audit.

4. DEDICATED PERSONAL RESPONSIBILITIES

Explicit Accountability for Every Task

Viranga Lakshan — Technical Lead

Primary Technical Owner

- Personally leads the PostgreSQL and Supabase audit.
- Investigates queries, indexes, locks, and connection pools.
- **Personally implements agreed database fixes and optimizations.**
- Leads database architecture decisions and technical sign-off.

Janith Kavishka — Backend/API & Load-Testing Eng.

Primary Load-Testing Owner

- Leads load-test planning and execution.
- Develops k6 and Locust test scripts.
- Generates synthetic test data for 100 restaurants and 100 salons.
- Executes tests against staging and supports API investigation.

Lithumitha Ravikumar — Project Coordination

Primary Coordination Owner

- Coordinates project requirements, timelines, and milestones.
- Manages client communication and status reporting.
- Prepares documentation and test deliverables; tracks progress.
- *Primary role is project management and coordination, not primary database engineering.*

5. ENTERPRISE CASE STUDIES & EVIDENCE (FULL DETAILS IN PROPOSAL)

Proven High-Concurrency Architectures Supporting Millions of Daily Transactions

Case Study 01: Mobile Field Platform (MFCS) — 1M+ DAILY RECEIPTS

High-concurrency mobile settlement network across **400+ nationwide sites** (Singer retail platform) processing **10+ Lakhs (1,000,000+) receipts daily**:

- **0.00% Record Loss:** Strict ACID boundaries with zero lost payments.
- **0 Database Locks / Deadlocks:** Wait times dropped from 1,450ms to 45ms.
- **p95 End-of-Day Sync Latency:** Dropped from **3,200 ms to 480 ms** (-85.0%).
- **Database CPU Load:** Slashed from **92% to 38%** during peak upload rush.

Case Study 02: Multi-Outlet Retail POS — 350 OUTLETS • 1,050 DEVICES

Enterprise multi-outlet POS platform (Perera & Sons) supporting **350 commercial outlets and 1,050 active devices** on shared database:

- **p95 API Latency:** Reduced from **920 ms to 420 ms** (-54.3%).
- **Query Response Time:** Slashed from **450 ms to 120 ms** (-73.3%).
- **Database CPU Load:** Dropped from **85% to 45%** during peak meal rush.
- **Connection Pool Efficiency:** Capped connections from **450 to 250**.

Immediate Kickoff & Reference Verification Protocol
Ready to Start: Phase 1 starts immediately upon Upwork milestone funding ($1,000) and staging database access. *Client reference introductions may be arranged subject to client authorization and availability. See our comprehensive 11-page master proposal (YouZen_Database_Optimization_Proposal2.pdf) for complete architectural schematics and k6 benchmarking blueprints.*